System Diagrams

Stage 1: SOUND CAPTURE

- Components: Microphone x3
- The 3D triangle spaced array captures ambient sound.
- Output: Sends "Analog Audio Signals" to Stage 2.

Stage 2: SIGNAL

- Components: Pre-Amplifier & Analog-to-Digital Converter (ADC) Module
- Output: Sends "Multichannel Digital Audio" to Stage 3.

Stage 3: DIGITAL SIGNAL PROCESSING

- - Noise Filtering & Focus Engine: Isolates Target Voice / Mutes Background Noise
  - MUSIC Algorithm Processor: Wave Computation & Real time Analysis
  - 3D Mapping: X, Y, Z Direction Finding for easy use.
- Output The processed data splits and is sent to the components in Stage 4.

Stage 4: STORAGE & INTERFACE

- Data Storage: Saves Raw Audio & Options for Export or Analyses.
- Visualization Processor: Renders Real-time Directional Overlay or Waveform visulisation.

Stage 5: OUTPUTS

- Display: 3D directional arrow indicator and sound visualization on an integrated screen (receives data from Data Storage).
- Review & Analysis Export: PC/Software interface for post-capture data review (receives data from the Visualization Processor).

- camera
- microphones
- camera mount that swivels it around
- controller + UI
- data capture
- computer
- data line: how samples get to computer from mics
- data line: how UI get to visual

- housing of mic & camera
- housing of controls near user

